

REPLACEMENT / MAINTENANCE

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SECTION 3 - MAGNET RAMPDOWN (DECREASE TO ZERO)

WARNING!

MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON, OR THE HATCH IS OPENED IF A MOBILE VAN, BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5-3 OF THE INTRODUCTION (CRYOGEN SAFETY).

A SUPERCONDUCTING MAGNET AT FIELD IS AN ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY DURING A QUENCH AND CREATING A DC VOLTAGE OF 100V OR MORE ACROSS THE MAIN LEADS AND EXTENSIONS.

NEVER TOUCH THE MAIN LEAD EXTENSIONS SIMULTANEOUSLY WHEN THE MAGNET IS AT "FIELD" AS A FATAL SHOCK AND/OR BURNS MAY RESULT IF A QUENCH OCCURS.

ENSURE INPUT POWER TO MAIN POWER SUPPLY IS DISCONNECTED WHEN CONNECTING MAIN POWER LEADS.

SECURE MAGNET ROOM DOORS IN THE OPEN POSITION BEFORE CHANGING MAGNETIC FIELD.

Note

Both Main Power Supply and Shim Power Supply are needed for the following procedure.

3-1 PREPARATION FOR FIELD CHANGE

1. Set up the field monitoring equipment, Probe and Teslameter in accordance with Set Up and Calibration, Section 5.
2. Ensure that the Magnet Power Supply is installed, checked and adjusted in accordance with the Vendor Manual (Sections 1 through 4) supplied with the unit. See Vendor Manual for location and description of power supply controls.
3. Ensure that the input Power Cable for the Power Supply is disconnected,
4. Retrieve the Main Coil connection polarity and latest final ramping current In Table 8-4 of Functional Checks.
5. Connect the Main Coil Power Supply to the magnet by making all cable connections In accordance with Set Up and Calibration, Section 6-2. Ensure power supplies connected with same polarity as retrieved In Step 4.

Note

This section allows one to remove all shim currents to find the actual main field to document and use as a target parking frequency when re-ramping the magnet. This will enable parking the magnet at a more accurate frequency and prevent retuning of the rest of the system.

6. Connect the heater cable, shim cable, and power cable to the Shim Power Supply in accordance with Set Up and Calibration, Section 8-1, Superconducting Shim Coil Power Supply Connection.

3-1 PREPARATION FOR FIELD CHANGE (continued)



MAKE SURE SHIM LEADS ARE “FROSTED” AND THERE IS SUFFICIENT FLOW OF HELIUM COOLING GAS THROUGH LEAD BEFORE SETTING ANY SHIM COIL CURRENTS IN THE POWER SUPPLY, TO PREVENT SHIM LEAD DAMAGE.

MAKE SURE THE CRYOGEN LEVEL IN THE MAGNET IS AT LEAST 75% TO PREVENT DAMAGE TO THE CORRECTION COILS OR A POSSIBLE QUENCH DURING THE FOLLOWING STEPS.

MAKE SURE CURRENT AND VOLTAGE CONTROLS ARE SET TO ZERO (FULL CCW) BEFORE TURNING ON THE POWER SUPPLY TO PREVENT UNINTENTIONAL SHIM LEAD CURRENT.

7. Remove the shim lead cap and precool the shim leads with GHe generated by the ramp lead extension insertion into the magnet.
8. Switch on the Shim Power Supply main power.
9. Turn on both Transverse Switch Heaters to dump all Transverse Shim currents, turn off switch heaters,
10. Set Axial Shim Coil currents in power supply to match currents in Axial Shims. See Table 9-4 in Set Up and Calibration,
11. Turn on Axial Heater, verify heater current is 810+/-10mA Allow two minutes for the heater to drive the switches resistive then slowly reduce all Axial Shim currents to zero,
12. Check the frequency reading on the Teslometer to make sure the Correction Coils are stable (i.e. there is no more than a 20 Hz change in total magnetic field over a two minute period).
13. Once the field is stable (see Step 12), record main field strength for reference when ramping magnet to operating conditions later, and turn off the Switch Heaters and allow the heater to cool.
Gauss
14. Switch off the Shim Power Supply main power.
15. Disconnect the Input Power Cable from the Shim Power Supply and connect it to the Main Power Supply.

3-2 RESISTANCE CHECKS

1. Ensure 'CURRENT CONTROL' and 'VOLTAGE CONTROL' on the Main Power Supply are off (full CCW).
Turn on Power Supply Input Power.
2. Turn on Axial Shim Heaters and observe current rise in ammeter (810 ± 10 mA) to ensure circuit continuity.
Ensure Main Heater Switch is off.

WARNING!

AXIAL SHIM SWITCH HEATERS MUST REMAIN ON DURING THE ENTIRE RAMPING PROCESS TO PREVENT IRREPARABLE SHIM COIL DAMAGE AND MAGNET QUENCH DURING RAMPING. THE POWER SUPPLY WILL NOT PASS CURRENT IN THE MAIN POWER LEAD CIRCUIT WITH THE AXIAL SHIM HEATERS OFF.

3. Connect a Digital Volt Meter (DVM) to the end of the Voltage Sense Leads terminated at the power supply.
4. Set "VOLTAGE CONTROL" on power supply to maximum (full CW).
5. Observe the Main Power Supply Ammeter and slowly turn the "CURRENT CONTROL" (CW) to pass 500A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
6. Record the voltage reading on the (DVM) _____ mV.

Note

A voltage reading less than 150 mV indicates acceptable internal contact resistance of the Lead Extensions. If the voltage reading is greater than 150 mV, turn "CURRENT CONTROL" off (full CCW), turn off Input Power and re-seat the Lead Extensions on the top of the magnet. Repeat Steps 1 through 7 when the Lead Extensions are re-seated. Repeated readings over 150 mV indicate the need to replace the Lead Extension Contact Bands. Refer to Replacement/Maintenance, Section 11.

Upon passing the internal resistance check, continue with Step 8.

7. Set the Power Supply Voltmeter to read voltage across the Power Supply Output Lugs.
8. Gradually increase the "CURRENT CONTROL" (CW) to pass 735A through the Main Power Leads, Lead Extensions and persistent Main Switch while observing the Power Supply Voltmeter. If the voltage exceeds 2.2V, discontinue the test.

3-2 RESISTANCE CHECKS (continued)**Note**

A voltage less than 2.2V at 735A indicates acceptable system resistance. If the voltage exceeds 2.2V during the test, turn the "CURRENT CONTROL" off (full CCW), turn off input power and check/tighten the bolts securing the Main Power Cables to the PowerSupply and Lead Extensions. Turn on power supply and repeat Step 9, after tightening Main Power Cable Connections.

9. Record the frequency displayed on the Teslameter while the 735A current is flowing through the switch.

Frequency (Hz) _____

Note

This is the target frequency for when the magnet is reramped.

Note

If a different frequency is desired, a delta of this frequency can be made. Refer to Set Up and Calibration, Section 11, Field Adjustment After Shimming, Step 1.

10. Upon passing Step 9, turn the "CURRENT CONTROL" off (full CCW) and continue with the Rampdown procedure.

3-3 MAGNET RAMPDOWN (DECREASE TO ZERO)

If a Quench OCCUR during change of magnetic field, Immediately turn "VOLTAGE CONTROL" and "CURRENT CONTROL" to zero.



ENSURE THAT THE CONNECTION POLARITY AND CURRENT SET ARE THE SAME AS THE LAST RECORD IN TABLE 8-4 OF FUNCTIONAL CHECKS. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.

3-3 MAGNET RAMPDOWN (DECREASE TO ZERO) (continued)

1. Set "CURRENT CONTROL" to pass the same current as retrieved in Replacement Maintenance, Section 3-1, Step 4, through the persistent Main Switch.
2. Turn on the Main Switch Heater. Leave the Axial Shim Switch Heater Supply on.
3. Allow approximately 1 minute for the Main Switch to go normal.
4. When the Main Switch is normal, gradually turn the "CURRENT CONTROL" to zero (over to a 1 minute period).
5. Turn the "VOLTAGE CONTROL" to zero (full CCW).

Note

When the "CURRENT CONTROL" is turned down the voltage across the magnet voltage taps will show a sudden decrease to (-1 to -2 V).

6. Observe voltmeter reading with voltmeter select switch in the "MAIN COIL" position. A 0.00X reading indicates that the magnet is fully discharged.
7. When magnet is fully discharged, turn off Axial Shim Heaters and Main Switch Heater.
8. Turn off Power Supply.
9. Disconnect input power from Power Supply.



Replace caps immediately after removing Main Power Lead Extensions to prevent ice build up inside Vertical Stack.

10. Disconnect the Main Power Leads on the top of the magnet and remove the Main Power Lead Extensions. Replace caps on Main Power Lead Extension Receptacles.